

Multiple Choice

1. **Answer:** B
Options: A) 1; B) 2; C) infinitely many; D) 0
Note: Correct option: B - 2
2. **Answer:** C
Options: A) subgroup; B) finite abelian group; C) infinite, non abelian group; D) infinite, abelian
Note: Correct option: C - infinite, non abelian group
3. **Answer:** C
Options: A) 2; B) 3; C) 4; D) 12
Note: Correct option: C - 4
4. **Answer:** D
Options: A) pair of points; B) circle; C) half-line; D) line
Note: Correct option: D - line
5. **Answer:** D
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: D - False, True

True / False

6. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Let $Q = (x, y)$. From the given information, $y^2 = 4 - 4x^2$. Therefore, PQ^2
 $= (x + 1)^2 + y^2$
 $= x^2 + 2x + 1 + 4 - 4x^2$
 $= -3x^2 + 2x + 5$
 $= -3\left(x - \frac{1}{3}\right)^2 + \frac{16}{3}$. This is maximized when $x = \frac{1}{3}$, and $d^2 = \frac{16}{3}$.
 Note: Students should show all steps in their mathematical solution.
12. **Answer:** II only. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
 Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.